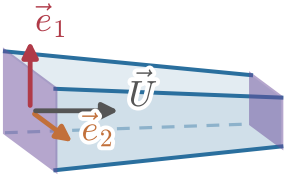
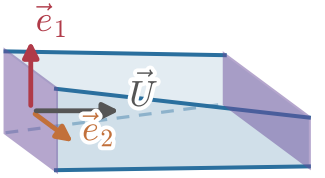
(a) $R_{\mu\nu}U^\mu U^\nu > 0$



$\kappa_1 > 0$ fecha
$\kappa_2 > 0$ fecha

as duas fecham:
o volume encolhe

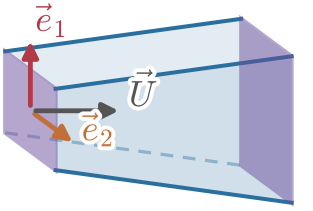
(b) $R_{\mu\nu}U^\mu U^\nu = 0$



$\kappa_1 > 0$ fecha
$\kappa_2 < 0$ abre

uma fecha o quanto a outra abre:
a área se conserva nesta ordem

(c) $R_{\mu\nu}U^\mu U^\nu < 0$



$\kappa_1 < 0$ abre
$\kappa_2 = 0$ fica

uma abre, a outra fica:
o volume cresce